Control admission | measurable effects, numerical limits and failed candidates

A Numerical quality and observed sensitivity

Metric	Observed	Admission / scope
Actual pressure	2781.7–2812.1 Pa	Table: 0.1–40000 Pa
Accepted overlap / R	0.0061776	≤ 0.01
Contact pass / R	0.0078519	≤ 0.1
Wall correction / R	0	≤ 0.1
COM dt difference	0.001308 nm	Observed, not a proof
COM cache difference	0.004666 nm	3 → 1.5 µm cache
Same-ID dt difference	6.416 µm	FAIL: tolerance 0.05 µm
Same-ID cache difference	3.041 µm	Particle layout unresolved
Shape Δ(X/Y)	0.004245	Numerical: 3.99e-07

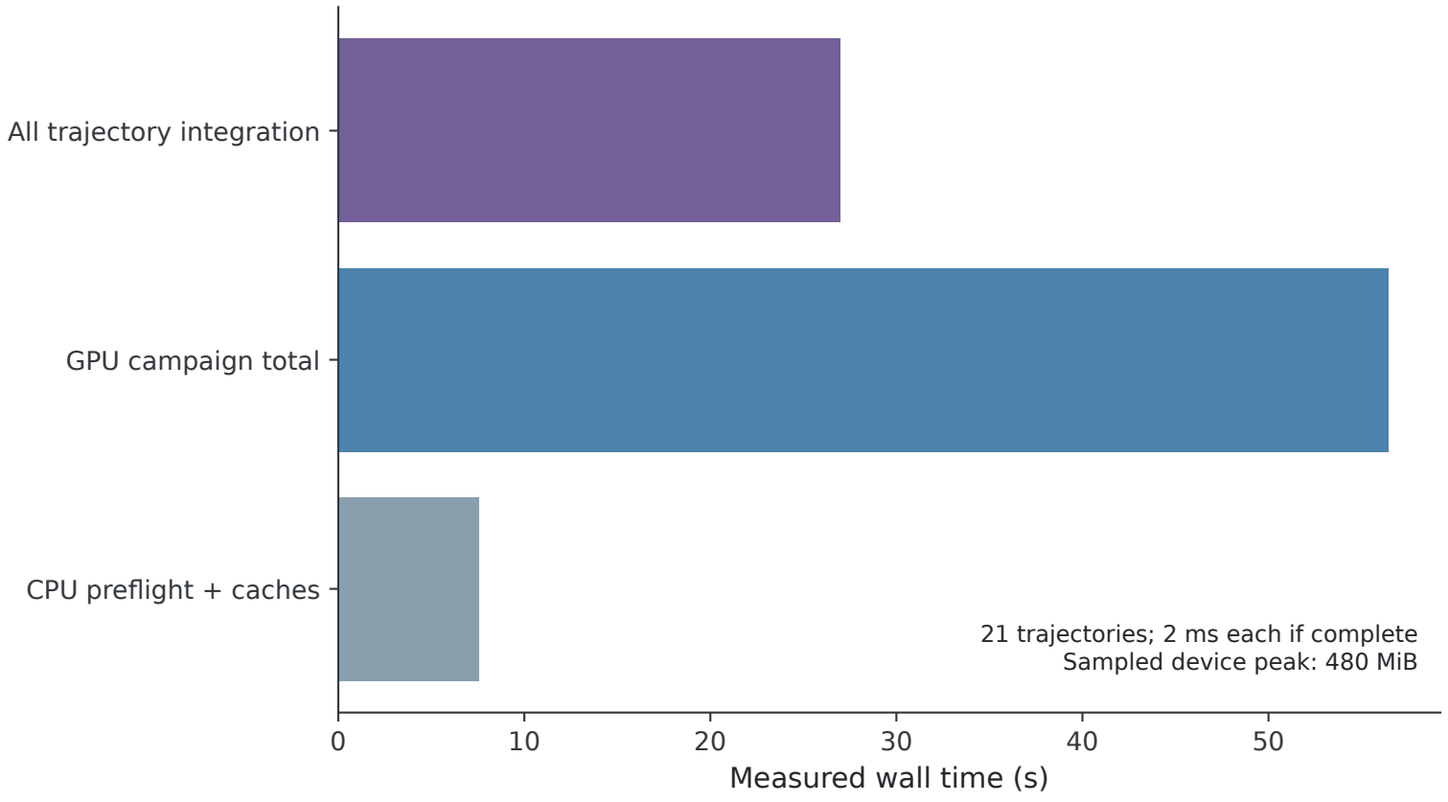
B Outcomes and retained failures

Position target reached: FALSE
Error reduction: 0.2002 µm
Shape target reached: FALSE
Completed / attempted: 20 / 21

NAV_STATIC_0: failed
Contact budget exceeded
Valid prefix and proposal retained.

Optimized particle layout: NOT CONVERGED.
Above/below-table tests: rejected without clipping.

C Bounded computation cost



D Scientific scope

HOMOGENEOUS_WATER_CONTROL_PROXY
EXPLORATORY / PHYSICS_UNCALIBRATED
training_ready = false

No PDMS reflection, free surface, CFD or streaming.
No experimental voltage/pressure calibration.
No independent signed Z translation demonstrated.
Instantaneous graph contacts remain time-sensitive.
Dense-contact physics and fusion are not validated.